TIC 232421797

Eclipsing binary candidate dossier - Star Vetter

CLEAN

```
BLS EPHEMERIS
  Period       6.192587 d
  Epoch (t0)   1468.9620 BTJD
  Duration     0.1700 d
  Depth        1.333 %  (13,334 ppm)
  SNR          134.5

INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     4  (gate >= 3)
  Robust depth ratio    1.03  (gate >= 0.50)
  Robust depth sigma    48.4  (gate >= 7.0)
  Odd/even z-score      2.23  (consistent)
  Secondary eclipse     not detected
```

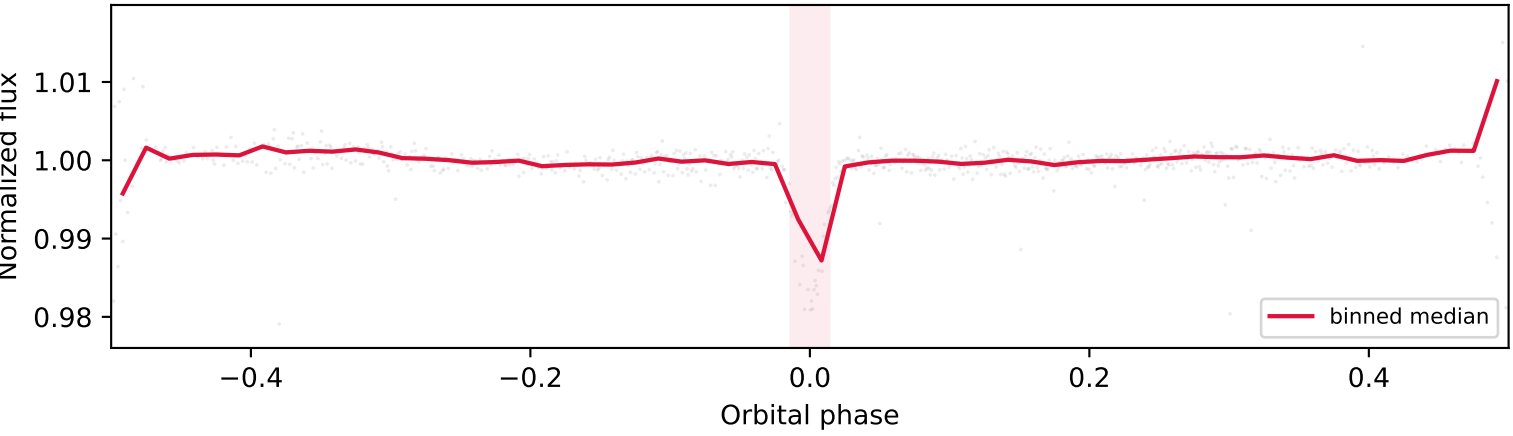
```
CATALOG CROSS-MATCH (novelty)

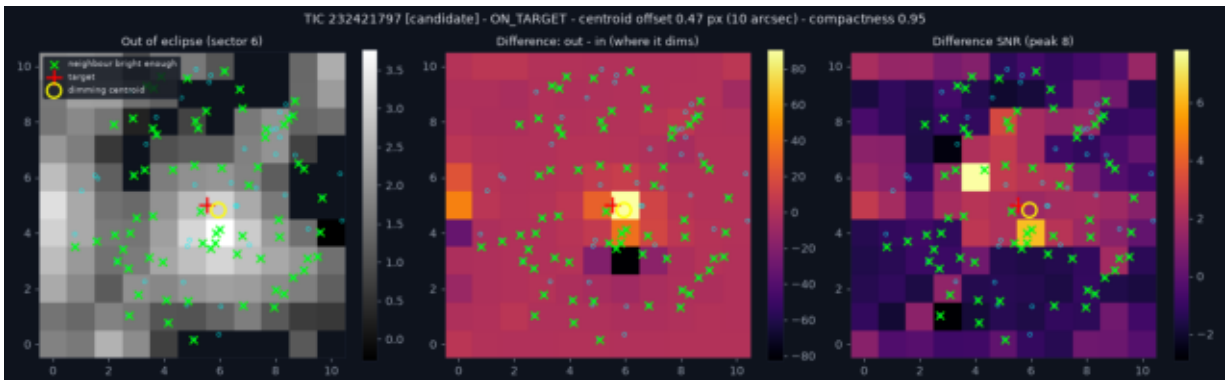
  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 6    light-curve dip in this sector: 50.2 sigma    t0 refined by -0.0025 phase
  Difference-image peak SNR   7.83   (gate >= 5.0)      Compactness   0.95   (gate >= 0.50)
  Dimming centroid offset     0.47 px (10 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 712310073
    0.65 px away, dTmag +3.91 - farther from the dimming than the target
```

Phase-folded light curve (shaded = BLS eclipse window)





No caveats - clean detection on every check.